

CIS 613: Metamorphic Testing

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How many pine needles are in this tree?

I have a method of counting.

I can tell you that there are
2,357,892 pine needles in this tree.

How can you tell if I am **correct**?

Could you create a **test
procedure**?

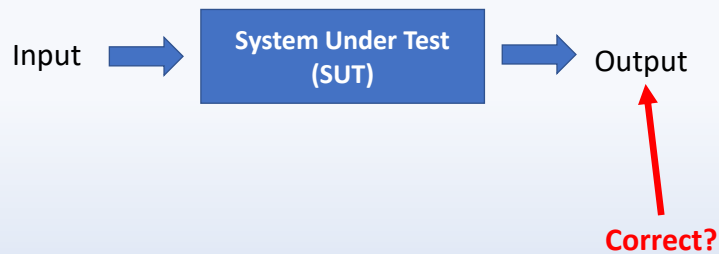


https://oregonstate.edu/trees/conifer_genera/spp/pine_spp.html

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Test Oracle

Test Oracle: Mechanism to decide whether a test output is correct or not.



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Test Oracle

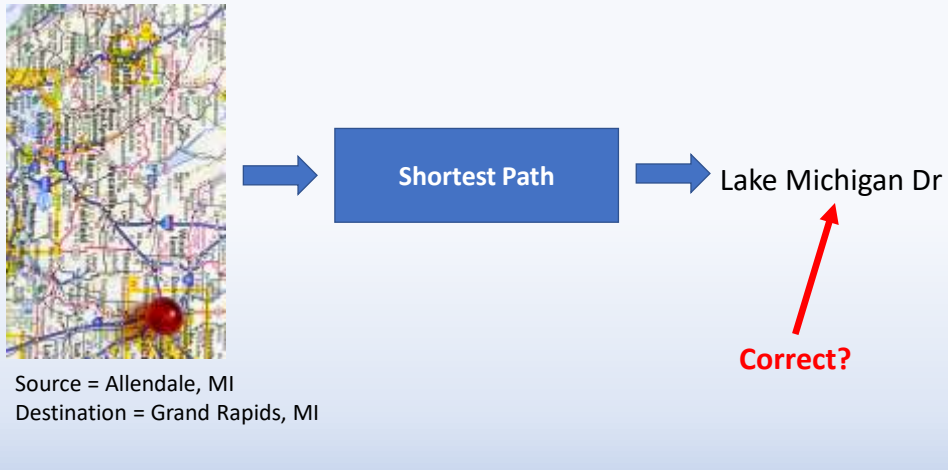
Test Oracle Problem: Sometimes it just isn't feasible to decide if test output is correct.

This is especially true in:

- Scientific Calculations
- Artificial Intelligence
- Simulation and Modeling

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The Oracle Problem - Examples



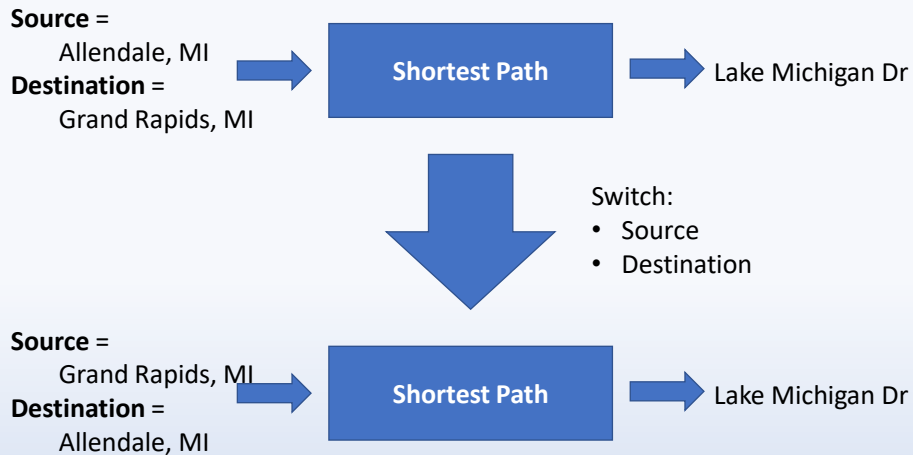
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The Oracle Problem - Examples



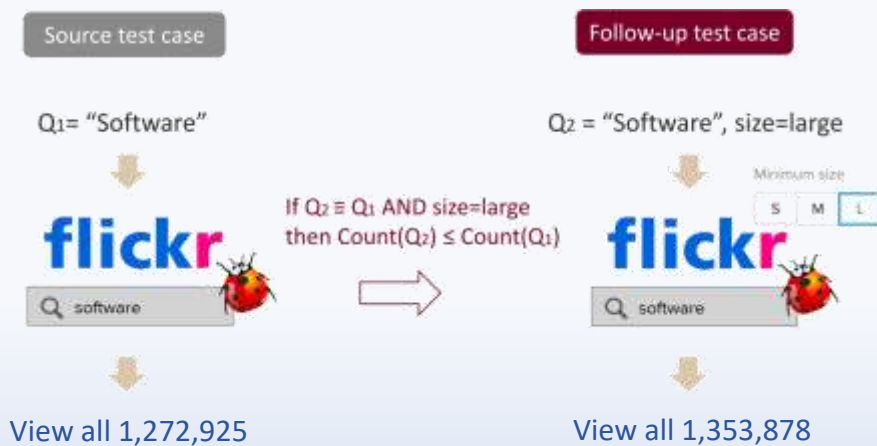
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Metamorphic Testing - Examples

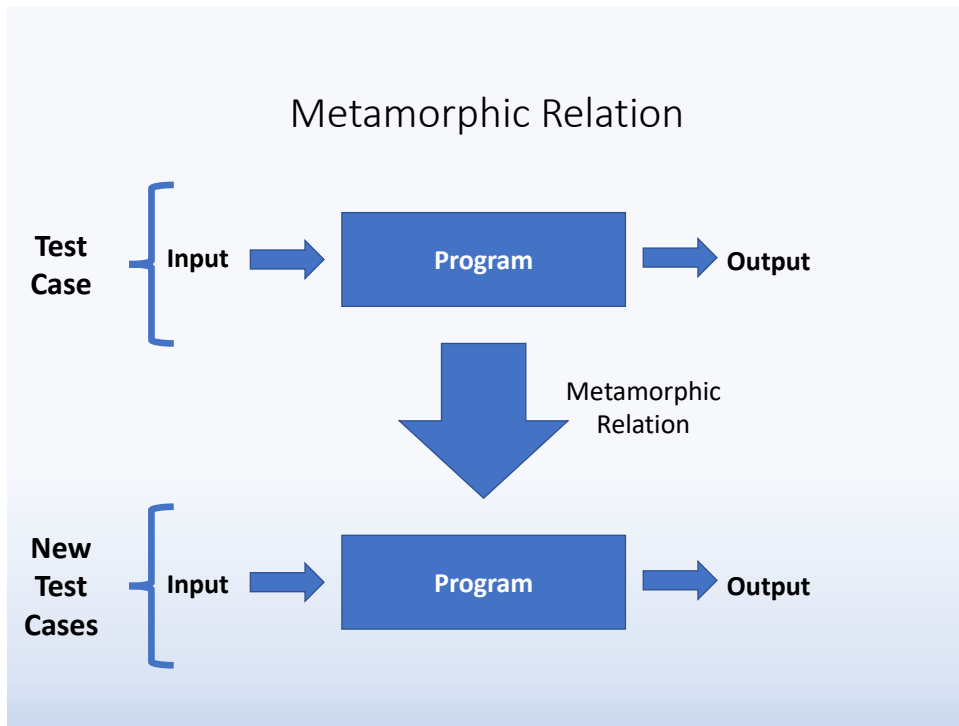


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Metamorphic Testing - Examples



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Reason for Metamorphic Testing

- One solution to the **oracle problem!**
- One one way to **generate test cases!**

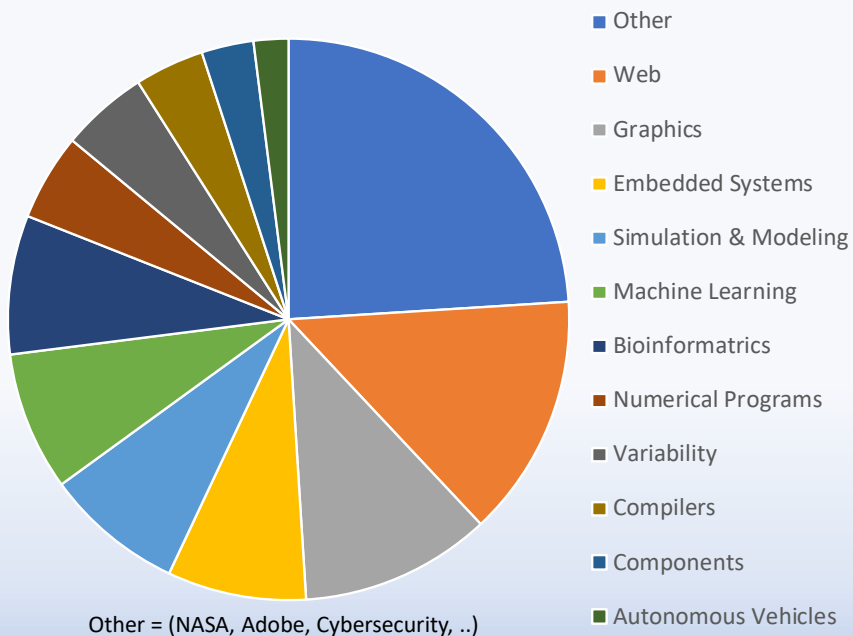
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How would we generate tests?

- Shortest Path:
 - Select a source and destination
 - Calculate route from source to destination
 - Switch source and destination
 - **Verify** the route is the same
- Flickr Searches:
 - Identify a keyword
 - Search for keyword to identify count of images
 - Add a restrictive filter
 - **Verify** the count is equal or lower

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Areas Metamorphic Testing is Used



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Guidelines for Metamorphic Testing

- Metamorphic testing requires good knowledge of the problem domain.

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Guidelines for Metamorphic Testing

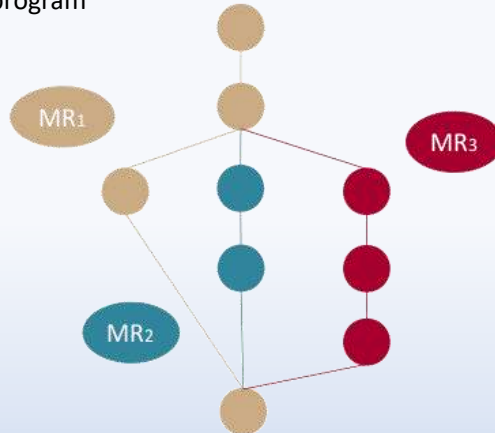
- Different metamorphic relations can have different fault-detection capability



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Guidelines for Metamorphic Testing

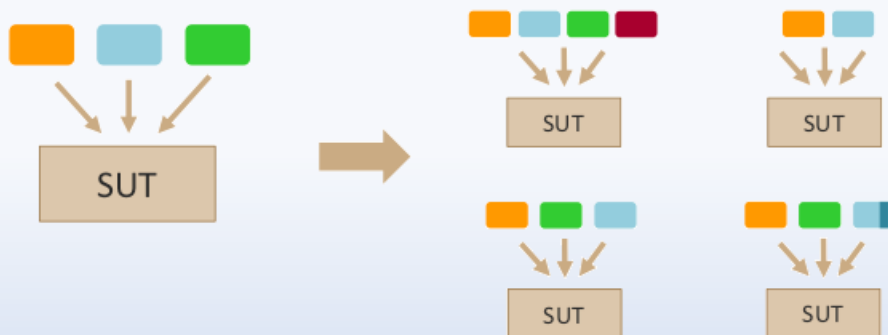
- Metamorphic relations should be diverse to they exercise different parts of the program



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Guidelines for Metamorphic Testing

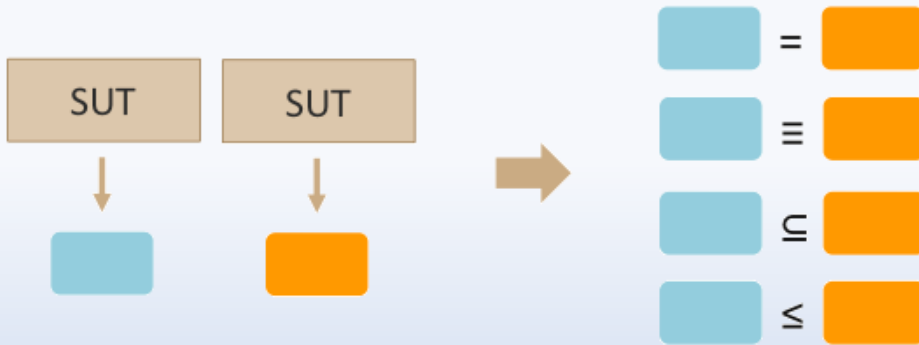
- Two common approaches for the construction of metamorphic relations are **input-driven** and output driven.



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Guidelines for Metamorphic Testing

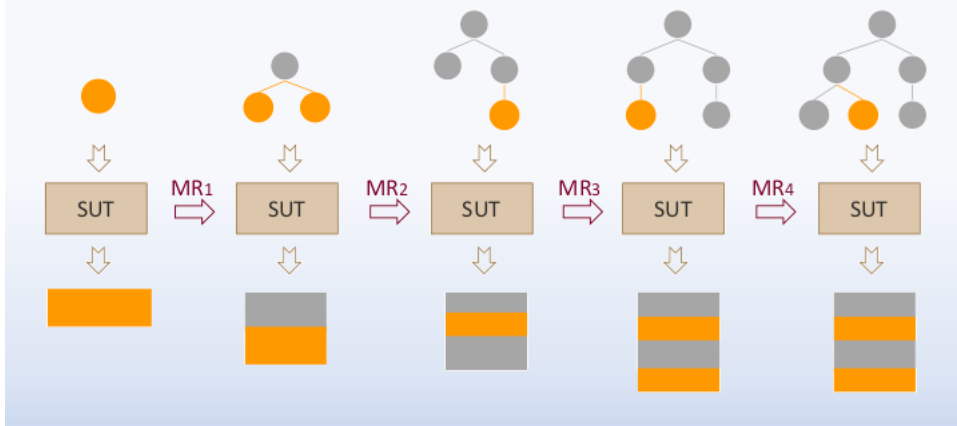
- Two common approaches for the construction of metamorphic relations are input-driven and **output driven**.



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Guidelines for Metamorphic Testing

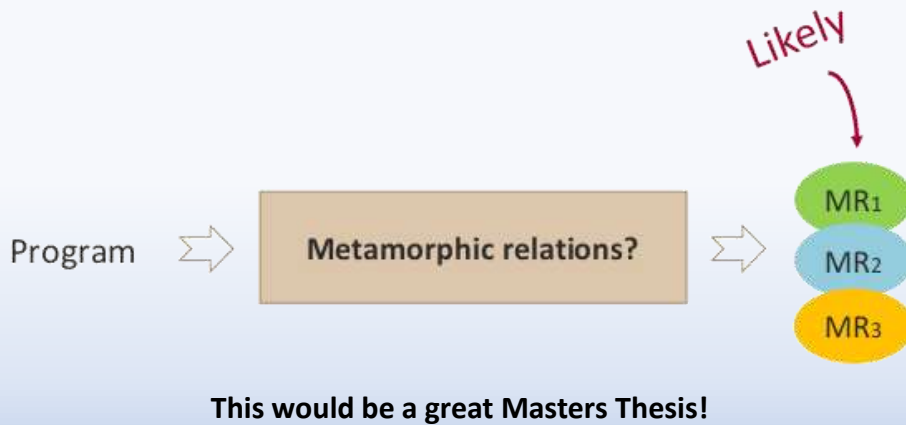
- Metamorphic relations can be combined



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Guidelines for Metamorphic Testing

- The automated discovery of metamorphism relations seems feasible in certain domains



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In Class Assignment!

- Identify metamorphic relations for the Shortest Path example!
- Your solution should define
 - The original tests case,
 - The metamorphic relationship, and
 - Updated tests cases.

Shortest Path accepts a graph (or map) and an origin (i.e., source) and destination. The shortest path algorithm returns the length *and* the path.

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In Class Assignment!

- Identify metamorphic relations for the NextDate example!
- Your solution should define
 - The original tests case,
 - The metamorphic relationship, and
 - Updated tests cases.

NextDate is a function of three variables: month, date, and year. It returns the date of the day after the input date. The month, date, and year variables have integer values subject to these conditions:

- c1. $1 \leq \text{month} \leq 12$
- c2. $1 \leq \text{day} \leq 31$
- c3. $1812 \leq \text{year} \leq 2012$

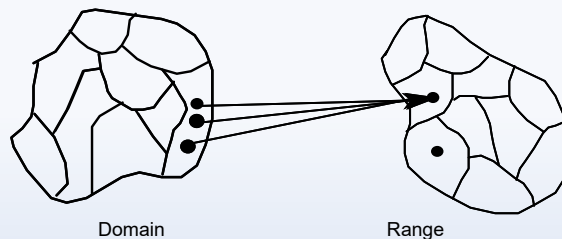
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Equivalence Partitions vs Metamorphic Relations?

Define a relation R on the input domain D as:

$$\forall x, y \in D, xRy \text{ iff } F(x) = F(y),$$

where F is the program function.



Equivalence Relations (e.g., R) treat inputs the *same*.

How do equivalence relations and metamorphic relations relate?

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Discussion

- What other problems would work well for Metamorphic testing?
- What are the downsides of Metamorphic testing?
- Is Metamorphic testing useful? Is it useful for “normal” businesses?

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What now?

An Assignment!

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